

BE511: Biomedical Instrumentation

Lab 1: Thermistors as Resistive Sensors

Collecting temperature data with thermistors

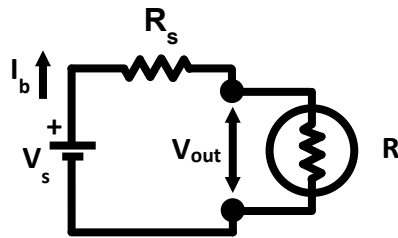
In this lab you will design, build, test, and characterize a thermistor-based temperature device that uses a **two-terminal bias circuit**. These types of circuits provide one of the simplest sensing schemes, which leverage a voltage measurement across a resistive sensor to determine the sensor resistance and thereby calculate a quantity of interest. In today's lab, that quantity is temperature. We will leverage the use of the National Instruments (NI) ELVIS Prototyping board together with the NI LabVIEW programming environment to measure, collect, and visualize the signal from our sensor. We will make use of the following items to build our temperature device.

Materials

- 1 x ELVIS Board
- 1 x Cantherm MF52C1103F3380 Thermistor
- Resistors (values to be determined)
- Jumper wires
- 2 x alligator clip to Dupont wire
- 1 x Hot plate
- 1 x Beaker
- LabVIEW
- MATLAB
- Microsoft Excel
- Banana cables with test leads
- Mercury thermometer

Introduction

Thermistors are a prime example of a resistive sensor. Resistive sensors are passive transducers that can convert a physical signal or quantity (like temperature) to a change in resistance through the sensor. To take advantage of this property, we can incorporate these devices into a simple two-terminal bias circuit, which allows us to control the maximum current through our system while also measuring the potential difference across the sensor. Using simple DC circuit analysis, we can utilize this potential difference and known quantities in the rest of the circuit to calculate the sensor resistance. The figure below provides a schematic of the circuit and an expression for the voltage across the resistive sensor in terms of the chosen source voltage (V_s), source resistance (R_s), and any instantaneous bias current (I_b).



Two-terminal bias circuit

$$V_{\text{out}} = -I_b R_s + V_s$$

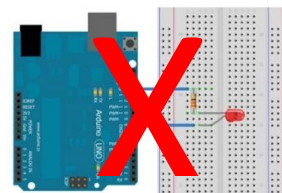
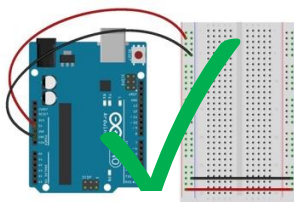
The relationship above is known as the **load line**, as it represents a line that describes all the possible currents and output voltages that can be observed across the load (the sensor). When designing a circuit, it is important to analyze the load line to ensure that there will never be a combination of voltage and current that could exceed the maximum power rating of the sensor, or any other electrical component used in the circuit. It is also important to ensure that no combination of voltage and current will exceed a power threshold that may lead to unacceptable measurement error. This will become clearer as we work with our thermistor in this lab.

In this lab, we will use a basic load line analysis as a starting point for designing and build our own two-terminal bias circuit for making safe and accurate temperature measurements. Our goals for today include:

- Learning basic usage of the NI ELVIS prototyping board and LabVIEW
- Constructing a simple temperature sensing circuit
- Exploring conditions for different load lines and implications of choosing one over another
- Characterizing and analyzing the behavior of a thermistor under different environmental conditions

Tips for Wiring and Using a Breadboard

- Use the shortest wires possible.
- Keep your board organized. Today's circuit is simple, but the more you practice good organization of your breadboards, the quicker you will be able to troubleshoot wiring problems later in the semester when we get to more complicated circuits.
- Power should be connected to the rails. All components that require power should draw from these rails, and never directly from the voltage source.



Part A: Determining Component Values

Before we can start building, we need to decide what values of V_s and R_s are suitable for our circuit. This will be driven by two factors: 1) The limitations of our power source – in this lab, we will use the variable voltage supply of our ELVIS unit to power the circuit; and 2) the maximum power rating of the thermistor. We must ensure that the power dissipated at any given time in the circuit is always less than this maximum power rating of our sensor.

1. Look up the Cantherm MF52C1103F3380 data sheet. **What is the maximum power rating of the thermistor?**
2. Using MATLAB, **plot a maximum power line** that sweeps voltages from 2-12V in increments of 2V. Plot voltage along the y-axis, and current along the x-axis.
3. We will use a supply voltage of +10 V to power our circuit. Given this value as V_s , calculate a table of values for R_s that correspond to each of the current values calculated above treated as maximum bias currents (think why that can be the case and how to use the load line equation to calculate R_s under max bias current conditions). **Plot load lines for each combination of V_s and R_s** by sweeping I_b from 0 to largest current value calculated above, all on a single plot and include the maximum power line. **Which load lines are not safe to use?**
4. Keeping in mind that the thermistor self-heating effect is minimized when less electrical power is dissipated, **which combination of V_s and R_s should we use for our circuit?** Also keep in mind the general IV curve of a thermistor and how bias currents influence the ohmic nature of thermistors.
5. Show your final numbers to the TA before continuing.

Part B: Characterizing parts

Before building your circuit, it is always important to characterize and record the actual values for your circuit components. Some components can have manufacturing tolerances that are $\pm 10\%$ of their nominal value, which can have a dramatic impact on the performance of your circuit. Knowing this up front can save you a lot of time later when trying to understand why your circuit is not performing as you would expect based on the nominal component values.

1. Find a resistor that most closely matches your chosen value for R_s .
2. Make sure the main power switch on the back of the ELVIS board is off.
3. Connect the power cable of the ELVIS board and the USB cable from the board to your computer.
4. Flip on the main power switch for the board. You should see a dialog window appear that confirms that they NI ELVIS has been detected. Close this window.
5. Search **Measurements Live** in the search box on your taskbar (near the Start menu) and open it.
6. In the dialog that opens, click on Measure → Connect (USB) → Instruments → **Digital Multimeter**.
7. Use this tool to measure and **record** the resistance of your resistor using a pair of banana cables.
8. Use the same tool to measure and **record** the resistance of your thermistor and **record** the approximate ambient temperature of the room based on a reading from your mercury thermometer.

Part C: Building Your Circuit

1. Connect the positive voltage and ground pins from the ELVIS variable voltage supply to the rails of your breadboard.
6. Add your source resistor to the circuit.
7. Connect an alligator-to-dupont pins to each lead of the thermistor. This will provide some more slack in your system, so that you can move your thermistor further away from the board.
8. Add the extended thermistor in series with the source resistor in your circuit.
9. Connect the A/AI0 analog input channel of the ELVIS board to positive terminal of the thermistor. Connect the A/AGND analog input channel to ground.
10. Open LabVIEW and choose LabVIEW for NI EVLIS III.
11. Create a new project and under Templates click NI ELVIS III → **NI ELVIS III Project** to use as the template. Name your project, select a project root path, and click Finish. Right click your EVLIS unit in the Project Explorer window and select Connect. Ensure Deployment Status says “Deployment completed successfully” and click Close. Right click your EVLIS unit again and select New → VI.
12. Two windows will now appear on your monitor. One is your block diagram, and the other is the front panel. The front panel represents the main user interface (UI) for your LabVIEW program. You can design and use controls on this front panel to interact with your program. The block diagram represents the business logic of your program. This is what controls the flow of data in your program and is where you would implement things like conditional statements, loops, and calculations. You may find it helpful to add “comments” to portions of your block diagram as you build it up, which you can do by double clicking in an empty area in the block diagram and type your comment (to move comment around in block diagram, click and drag with mouse and release to place).
13. Click on the block diagram and add a while loop by going to the Functions Panel and clicking on **Programming > Structures > While Loop**. Draw a rectangle on the block diagram to add the loop.
14. In the same Functions Panel, drag an Analog Input object into the while loop from **Academic I/O > Control I/O > Analog Input**. A dialog window will automatically open.
15. Configure the I/O mode as “Analog input (1 sample)”, the analog input Channel as “A/AI0”, and the Range as +/- 10V. This sets up the Analog Input so that it captures just a single value from the ELVIS board. Click OK.
16. Add a 100 ms delay inside the loop by dragging in the Wait (ms) object from **Programming > Timing**. Hover over the left side of the block and right click on the blue handle. Click on **Create > Constant** and enter 100 in the box that appears. This will serve as a placeholder for now.
17. Go to the Front Panel window.
18. In the Controls panel, drag a Waveform Chart onto the Front Panel from **Modern > Graph > Waveform Chart**.
19. From the same Controls panel, drag in a Stop button from **Modern > Boolean > Stop Button**. We will use this to stop the program whenever we’re ready to do so.
20. Go back to the Block Diagram window. You will now see objects there for the Waveform Chart and Stop button. Make sure these are inside the while loop.

21. Drag a wire from the right side of the Stop button to the red stop icon in the bottom right of the While loop. A signal from this Stop button is what will tell the while loop to stop.
22. Drag a wire from A/AI0 on the right of the Analog Input to the left orange connection of the Waveform Chart. This will stream data collected from the Analog Input to the chart that you placed on the Front Panel.
23. Make sure you save your work if you haven't already.
24. Go back to the Front Panel and click on the **Run** button just below the File menu. It looks like a single arrow pointing to the right. You should now see voltage readings across the thermistor streaming to the chart in the front panel.
25. Click on the Stop button (the one you put in your front panel, NOT the stop symbol near the run arrow) to stop the VI.
26. In the Front Panel, drag in a numeric indicator to display the instantaneous voltage reading from the ELVIS board. This can be found under **Modern > Numeric > Numeric Indicator**. Double-click the label where it says "Numeric" and label the control "Voltage."
27. We're also going to add a slider to allow us to change the wait time within the loop so that we can control our sampling period. Drag in a Horizontal Pointer Slider from **Modern > Numeric > Horizontal Pointer Slider**. Label the slider "Delay (ms)".
28. Right-click the Horizontal Pointer Slider and click on Properties. Navigate to the **Scale** tab and enter 10 to 2000 for the minimum and maximum values, respectively.
29. Return to the Block Diagram and make sure the new controls are inside the While loop.
30. Delete the numeric constant connected to the Wait object (wristwatch symbol) and instead connect the output of the Horizontal Pointer Slider [Delay (ms)] to it.
31. Now connect the input to Numeric Indicator (Voltage) to the data wire coming from the A/AI0 of the Analog Input.
32. On your ELVIS device, switch on the APPLICATION BOARD POWER. Launch Measurements Live, click Measure → Connect (make sure your ELVIS device is recognized and selected), and under Instruments open the Variable Power Supply (VPS). Set Supply + to 10 V, then hit Run.
33. Return to your Front Panel and run the VI.
34. Confirm your circuit is functioning as you expect by calculating the temperature based on the voltage you are seeing in the VI. You will need to look up and use the thermistor's β and R_{25} values and calculate the thermistor resistance using the load line equation and Ohm's law. Assuming no change in the mercury thermometer temperature from Part B, **how do the calculated thermistor resistance and temperature values from the measured voltage compare to the recorded thermistor resistance and mercury thermometer readings from Part B?**
35. Update your block diagram to automatically calculate the temperature based on the equations we discussed in class. You can find the appropriate controls in the Block Diagram under the **Mathematics > Numeric section** of the Functions panel. In updating your block diagram with this calculation, you may find it helpful to create **Numeric Controls** and **Numeric Indicators** on the front panel for aspects such as resistances, voltages, currents, temperatures, constants, etc. to be able to control inputs and display intermediary outputs while the VI runs.
36. Keep the plot of the thermistor voltage reading and change the title of the plot accordingly. Change the y axis label to Voltage (V). Find the corresponding display item for the plot in the block diagram

and right click it to change its properties. Under the Display Format tab, select Time (X-Axis) from the dropdown, set Type to Floating point, set Digits to 6, set Precision Type to Digits of precision, and check Hide trailing zeros. Under the Scales tab, select Time (X-Axis) from the dropdown, set Name = Time (secs), check Autoscale, and under Scaling Factors set the Multiplier to the delay value on the ms slider converted to seconds. Click OK. Right click the plot display area on the front panel, select Data Operations -> Clear Chart to reset the plot. Do all of this for another plot display for the thermistor temperature based on your calculations in the block diagram.

37. Add another Numeric Indicator that displays the calculated temperature on the Front Panel in addition to the voltage.
38. Show the real-time output of the final VI to the TA.
39. **Take a screenshot of both the front panel and block diagram.**

Part D: Temperature Response

Let's now look at **how the thermistor responds to gradual changes** in temperature and **evaluate its accuracy** against a mercury thermometer as a control.

1. Fill a glass beaker with water.
2. Clear a space near your ELVIS board for your hot plate and water. Make sure that wherever you place your water, the tip of the thermistor can reach the surface. Place both your thermistor and mercury thermometer into the water.
3. In an excel spreadsheet or MATLAB, you will **record the mercury thermometer temperature and thermistor voltage and temperature** at regular intervals of approximately 5°C until the water reaches 70°C. Think carefully about what sampling period to use (setting for delay slider) such that you can see changes in temperature in real time on the Front Panel while your VI runs (make sure Multiplier for Time axes matches delay slider). Once ready to record, turn on hot plate and run VI.
4. When the water reaches 70°C according to the mercury thermometer, adjust your hot plate setting to maintain the water temperature around 70°C for the next part.
5. Compare the overall thermistor response to that of the mercury thermometer in your lab report.

Part E: Source Resistance

Now that we have a functioning circuit and have observed the behavior of the thermistor with gradual changes in temperature, let's try to understand and characterize it better by looking at how it responds to rapid changes in temperature and changes in the source resistance, R_s .

1. In Part E, you will want to export the thermistor voltage and temperature data of each measurement, which you can do by right clicking the display area of the plots on the front panel and selecting Export -> Export Data to Excel. Only the data displayed on the plots will be exported, so make sure that for each measurement the entire data is displayed on the plots before exporting. Remember to also clear the plot displays between measurements.

2. Think carefully about what sampling period to use (setting for delay slider) such that your VI will acquire multiple data points during rapid (think step function like) changes in temperature. Make sure the Multiplier for the Time axes are set to the delay value of the ms slider in seconds.
3. With your thermistor still in the air and away from the 70°C water, start your VI to begin collecting data. Collect about 10s of data (or until there's no drift in the readouts), and then dunk the tip of your thermistor into the hot water. Watch your Waveform Chart and wait for the signal to stabilize. Once it has stabilized, remove the thermistor from the water and away from the beaker and wait for the signal to equilibrate again.
4. Stop the VI to export and save the data set. Double check that the data exported successfully. You may want to also do a quick plot of the data to ensure there are multiple data points during phases in which temperature/voltage rapidly change.
5. **Repeat to collect a total of 3 data sets.**
6. Add a second source resistor of the same value in series with the current source resistor.
7. **Measure the new combined resistance and record this value.** Make sure the circuit is not powered when making your measurement.
8. **Collect another three data sets with the new source resistance.**
9. Change your source resistance a third time, this time by reducing the overall source resistance (hint: you can put your existing resistors in parallel), but make sure that the new resistance keeps the total power below the maximum power line of the thermistor.
10. **Measure the new combined resistance and record this value.** Make sure the circuit is not powered when making your measurement.
11. **Collect three more data sets with this final source resistance.**
12. Feel free to use MATLAB or any equivalent computational tool (NOT Excel) for data analysis.

Part F: Data Analysis

Except for steps 1(i) and 2(i), you may perform the rest of this section outside of class. **The following analyses should be included in your final lab report (as well as any bold sections in the previous parts):**

1. From **Part D, step 3**, plot and analyze the following: (i) thermistor voltage vs thermometer temperature; (ii) thermistor resistance vs thermometer temperature, and on same axes *expected* thermistor resistance vs thermometer temperature; (iii) thermistor temperature vs thermometer temperature, and on the same axes the *ideal* thermistor temperature vs thermometer temperature relationship; (iv) residuals between temperature measured with your digital thermometer and temperature measured with the mercury thermometer vs the thermometer temperature. **What trends do you observe? Provide rationale for trends that you observe.** (Some questions to consider: Are plots (i) and (ii) linear? Explain why or why not? How well do your calculated temperature values agree with the thermometer temperatures? What is the mean residual value? Are the residuals of equal magnitude over the temperature range? Why do you think this is?)
2. From **Part E**, (i) plot the time data (temperature and voltage) from each resistor setup, then (ii) plot delta versions of the same time data by subtracting the first data point from itself and all data points following it. **What trends do you observe? Provide rationale for trends that you observe.**

3. **Calculate the average time constant with standard deviation for each resistor setup using the temperature vs time data for when you submerge the thermistor in water.** The time constant represents the amount of time it takes for the thermistor to change from its initial value right before being exposed to a step input change (like going from air to hot water) to $[(\text{final value} - \text{initial value}) * (1 - \frac{1}{e}) + \text{initial value}]$. **Mark the start and end points on the curves for calculating the time constants. Briefly illustrate why quantifying the time constant might be important through an example biomedical application involving temperature sensing.**
4. **What is the average time constant with standard deviation for each resistor setup using the temperature vs time data for when you take it out of the water and leave it in air (time between initial value and $(\text{initial value} - \text{final value}) * (1/e) + \text{final value}$)? Mark the start and end points on the curves for calculating the time constants. How do the dunk in water vs remove from water time constants compare to each other and to the time constant in the data sheet? Provide rationale for observed comparison among all the time constants. How would you make the thermistor response faster?**
5. **From Part E, what is the power dissipation of the thermistor for each resistor setup at the reference temperature of 25°C? Which resistor setup gives the lowest thermistor power dissipation? Is this expected? Why or why not?**
6. **From Part E, plot thermistor voltage vs thermistor resistance for each resistor setup. What do you observe about the magnitude and range of the output signal for each of the resistor setups? Provide rationale for what you observe.**
7. **Which resistor setup would you choose to use for a final sensing circuit and why?**

Part G: End of Lab

Show basic plots of steps 1(i) and 2(i) from Part F to your TA before leaving.